

Scaleform CLIK

Scaleform CLIK

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: 2.0
: 2010 8 19

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Scaleform CLIK User Guide (Scaleform CLIK)

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Scaleform® Common Lightweight Interface Kit (CLIK™)
CLIK User Guide

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2

[Getting Started with CLIK Buttons](#)
Scaleform CLIK
CLIK

Scaleform CLIK ActionScript™ 2.0 (AS) Scaleform 4.4 (UI)

PC (UI CPU)

CLIK 15)

(

:

UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	ProgressBar
RadioButton	UILoader
Label	ScrollingList
TextInput	TileList
TextArea	DropDownMenu
ScrollIndicator	Dialog
ScrollBar	NumericStepper

:

API

InputManager	Locale
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IdataProvider

EventDispatcher	Ilist
Delegate	IlistItemRenderer
Constraints	GameDataProvider

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK API Reference
CLIK User Guide
CLIK Flash Samples
CLIK Video Tutorials

Flash

Flash

gskinner.com

Scaleform

Scaleform CLIK

gskinner.com		Adobe
Flash Creative Suite® 4 (CS4)		
gskinner.com	Flash	1
gskinner.com		http://gskinner.com/blog

CLIK

CLIK

Flash Studio

:

C:\Program Files\Scaleform\GFx 4.4\Resources\AS2\CLIK\components\CLIK_Components fla

C

UI

UI

Microsoft Visio®

Flash UI

Scaleform CLIK

UI

Flash

Visio

Flash

CLIK

UI

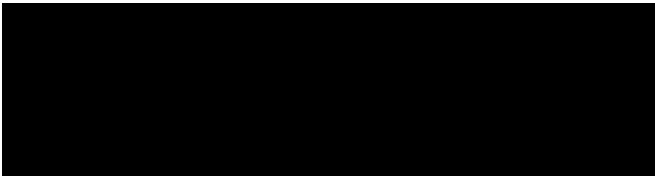
- :
- : Flash
- : (
-)
- Flash 4
- 4UI
- 4

Button CheckBox Label TextInput TextArea
CheckBox Button Label Button
TextInput TextArea textField

1: *Free Realms* –

2: *Free Realms* -

3: *Free Realms* -



4: *Crysis Warhead* -

TextInput

5: *Mercenaries 2* -

TextArea

6:

Button CLIK

(gfx.controls.Button)

Button

textField

ScrollBar

Button

Slider

Button

7: `AnimatedButton` `AnimatedToggleButton` `ToggleButton`

CLIK Button

Button.fla

ToggleButton AnimatedButton

AnimatedToggleButton

Button

CLIK

Tab

Shift+Tab ()
 Web CLIK
 Shift+Tab CLIK
 Flash Tab
 CLIK

Button Button Enter
 Button Button /

Xbox360 PS3
 CLIK A X Enter
 UI

CLIK Button MovieClip

- *textField* : () TextField
- *focusIndicator* : () MovieClip

"hide" 2
 over

MovieClip

MovieClip "show"

Button

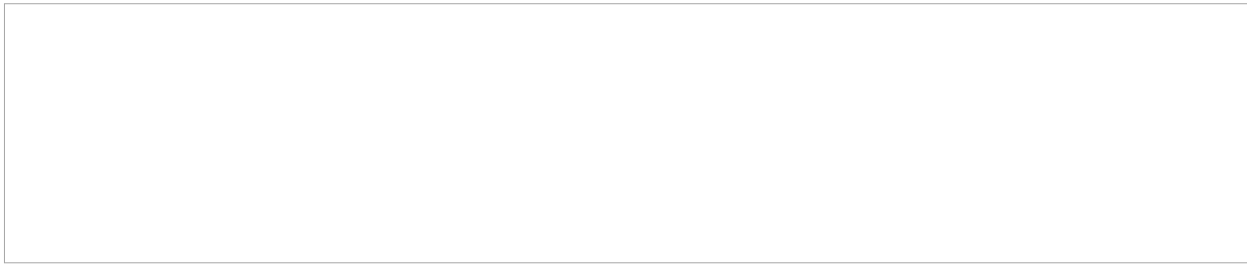
focusIndicator
 Button

CLIK Button

:

- *up*
- *over*
- *down*

- *disabled*



8: Button

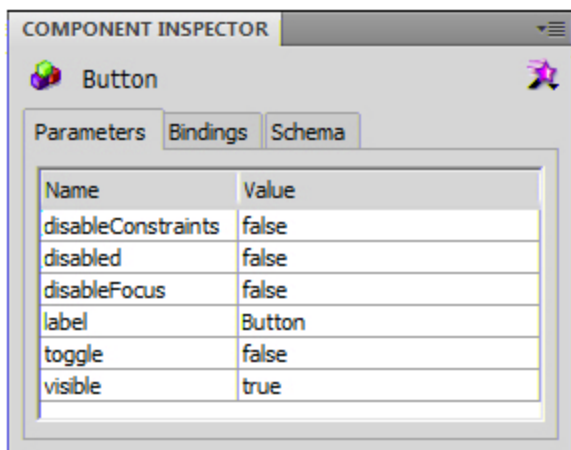
Flash

Button

Buttons

Getting Started with CLIK

Flash IDE [CS4 [Shift
+ F7 [] AS



9: CS4 [Button

Button :

disableConstraints Button

	constraints (	textField	)	constraints
	true			
	textField			
	textField			
	textField		/	
Disabled	true			
disableFocus				true
Label	Button			
Toggle	Button	(	)	true
		Button		
Visible	false			

Flash IDE CLIK SWF

CLIK

1 Object

- *type* :
- *target* :

Button

Show	visible	true
Hide	visible	false
focusIn		
focusOut		

Select	<code>selected</code> <code>selected : Button</code> <code>true</code>
stateChange	<code>state : Button</code> <code>"up" "over" "down"</code> Getting Started with CLIK Buttons
rollOver	<code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code>
rollOut	<code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code>
Press	<code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code>
doubleClick	<code>doubleClickEnabled</code> <code>true</code> <code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code>
Click	<code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code>
dragOver	<code>(</code> <code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code> <code>)</code>
dragOut	<code>(</code> <code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code> <code>)</code>
releaseOutside	<code>mouseIndex :</code> <code>(</code> <code>0 3</code> <code>)</code>

ActionScript

:

```
myButton.addEventListener("click", this, "onButtonPress");
function onButtonPress(event:Object) {
    //
}
```

'myButton'

"click"

onButtonPress

Object

(

'event'

)

Object

Button

:

```
import gfx.controls.Button;
attachMovie("Button", "btnInstanceName", someDepth, {_x:33, _y:262, ...});
```

CLIK Button

toggle

:

```
btn.toggle = true;
```

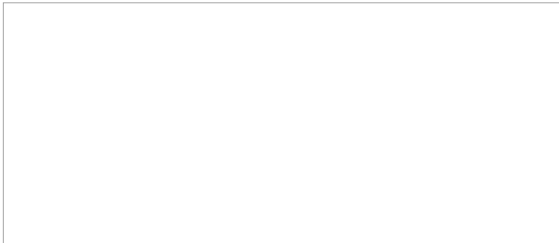
:

```
btn.doubleClickEnabled = true;
btn.addEventListener("doubleClick", this, "onDoubleClick");
function onDoubleClick(e:Object):Void {
    //
    !
}
```

click

:

```
btn.autoRepeat = true;
```



10: CheckBox

CheckBox (gfx.controls.CheckBox)
 Button

 CheckBoxes true/false ()
ToggleButton

toggle

CheckBox

CheckBox Button

/

CLIK CheckBox

MovieClip

:

- *textField* : () TextField
 - *focusIndicator* : () MovieClip
- MovieClip "hide" 2

MovieClip "show"

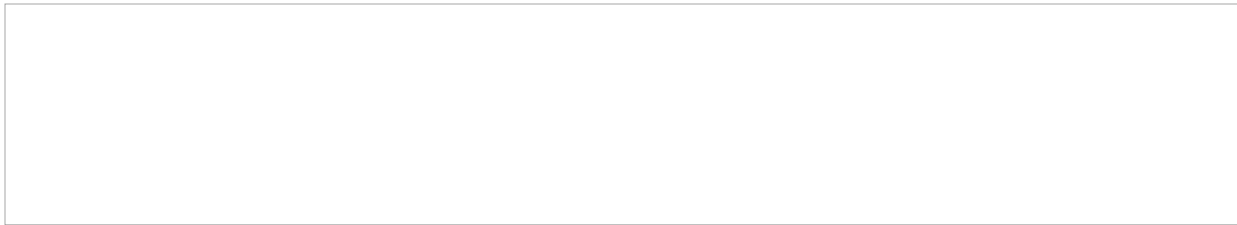
toggle

CheckBox

:

- *up*

- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*



11: CheckBox

CheckBox

Button
CheckBox

[Getting Started with CLIK Buttons](#)

CheckBox Button

toggle disableFocus

Button

Data	CheckBox
disableConstraints	Button
	constraints (true)
	textField
	constraints
	textField
	textField /
disabled	true
Label	Button

selected	true	CheckBox	(
Visible	false		)

1 Object

- *type :*
- *target :*

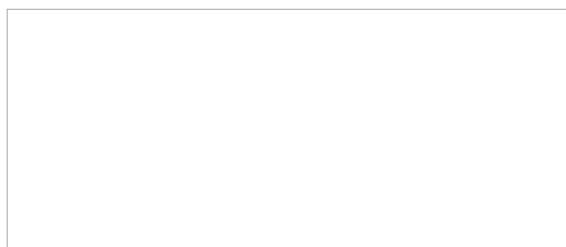
CheckBox

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
Select	selected	
	<i>selected</i> : Button	selected
stateChange	<i>state</i> : Button "up" "over" "down" Getting Started with CLIK Buttons	
rollOver	<i>mouseIndex</i> :	(0 3)
rollOut	<i>mouseIndex</i> :	(0 3)
Press	<i>mouseIndex</i> :	(0 3)
doubleClick	true <i>mouseIndex</i> :	<i>doubleClickEnabled</i> (0 3)
Click	<i>mouseIndex</i> :	

	(	0	3
dragOver	)	(	)
	<i>mouseIndex :</i>	(	)
	)	0	3
dragOut	(	)	)
	<i>mouseIndex :</i>	(	)
	)	0	3
releaseOutside	(	)	)
	<i>mouseIndex :</i>	(	)
	)	0	3

CheckBox

```
myCheckBox.addEventListener("select", this, "onCheckBoxToggle");
function onCheckBoxToggle(event:Object) {
    //
}
```



12: Label

CLIK Label (gfx.controls.Label) textField
 MovieClip
 Label textField
 Label textField
 Label
 textField
 Label
 textField

Label	MovieClip
-------	-----------

```

    textField
      textField
        Label
      textField

```

disabled
AS2

Label

Label
textField
textField

Label

textField

Label

Label

CLIK Label

MovieClip

•

•

- ***n r n c*** : TextField Label

CLIK Label

disabled

2

•

•

- *default*
- *disabled*

13: Label

Label :

Text	
Visible	false
disabled	true

1 Object

- *type* :
- *target* :

Label

Show	visible	true
Hide	visible	false
stateChange	Label <i>state</i> : "default"	Label "disabled"

Label :

```
myLabel.addEventListener("stateChange", this, "onStateChange");
function onStateChange(event:Object) {
    if (event.state == "default") {
        //
    }
}
```

Label HTML HTML textField
HTML Flash 8 :

```
lbl.htmlText = "<b>foo</b>bar";
```



14: TextInput

TextInput (gfx.controls.TextInput)
textField

textField
textField

Label

HTML

TextInput textField

TextInput

Label
textField
UI

TextInput

Label
TextInput
PC
textField

TextInput focused disabled
textField

TextInput
textField
AS2

TextInput
textField
TextInput

Flash Studio
TextInput

TextInput HTML textField

(Shift+) (Shift+Delete) (Ctrl+ Insert)
(Shift+ Insert)

TextInput
actAsButton inspectable

“default” “over” rollOver rollOut “over” “out”
 actAsButton
 TextInput CLIK

Text String
 0xAAAAAA
 TextFormat Text

TextInput
 textField

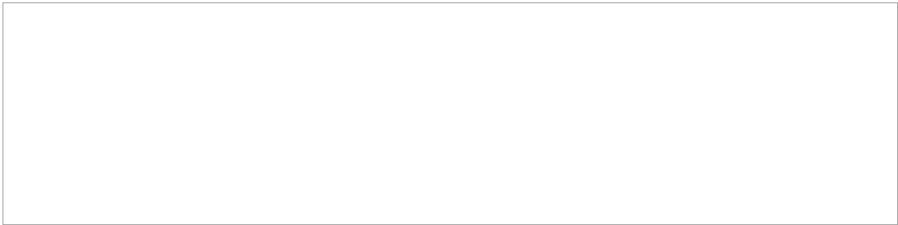
 textField

CLIK TextInput MovieClip :

- ***nrnc*** : textField

CLIK TextInput : focused disabled 3

- *default*
- *focused* textField
- *disabled*



15: TextInput

TextInput :

Text	textField		
Visible	false		
disabled	true		
editable	false	TextInput	
maxChars	textField		
password	true	'*'	textField
	textField		
	text		
defaultText	textField		
	defaultTextFormat		
actAsButton	true	TextInput	Button
		rollOver	rollOut
	TextInput		

1 Object

- *rs* :
- *n la n4*

TextInput

Show	visible	true
Hide	visible	false

focusIn		
focusOut		
textChange	textField	
rollOver	actAsButton	
	<i>mouseIndex:</i>	
	0	3
rollOut	actAsButton	
	<i>mouseIndex:</i>	
	0	3

textField :

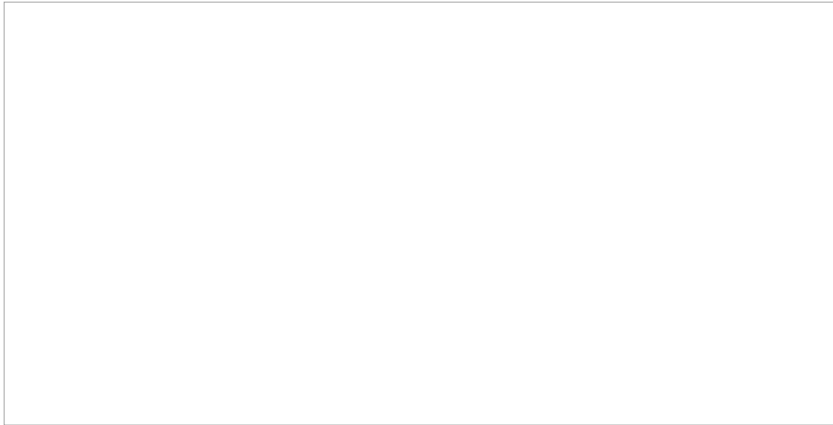
```
myTextInput.addEventListener("textChange", this, "onTextChange");
function onTextChange(event:Object) {
    //
}
```

:

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextInput HTML textField
HTML Flash 8 :

```
textInput.htmlText = "<b>foo</b>bar";
```



16: TextArea

TextArea (gfx.controls.TextArea) CLIK TextInput

ScrollBar
TextInput

TextInput TextArea

Label TextInput
textField

TextArea
HTML

textField
TextArea textField

textField ScrollIndicator ScrollBar
TextArea

TextArea textField

TextArea

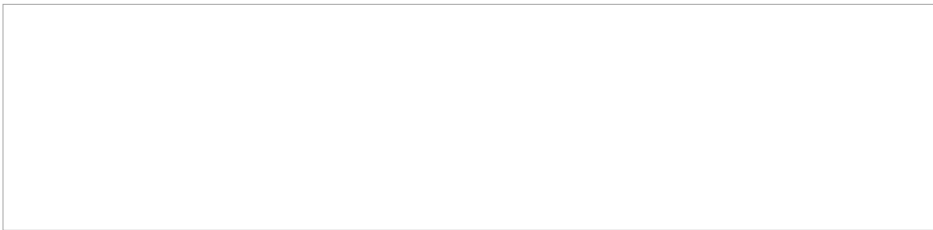
TextArea
textField

CLIK TextArea MovieClip :

- *textField* : TextField

focused disabled TextInput 3 TextArea :

- *on*
- *ion* textField
- *on*



17: TextArea

TextArea password TextInput CLIK ScrollBar 2.4 :

Text	textField	
Visible	false	
disabled	true	
editable	false	TextInput
maxChars		textField
scrollBar	CLIK ScrollBar ScrollBar TextArea	ID ()

[illegible]

1 Object

- n_5 :
- $n_{la} \ n_4$

TextArea

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
textChange	textField	
Scroll		
rollOver	actAsButton	
	<i>mouseIndex:</i>	
	0	3
rollOut	actAsButton	
	<i>mouseIndex:</i>	

TextArea :

```
myTextArea.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    //
}
```

:

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextArea
HTML

HTML
Flash 8

textField
:

```
textInput.htmlText = "<b>foo</b>bar";
```

RadioButton ButtonGroup

ButtonBar

ButtonGroup Button

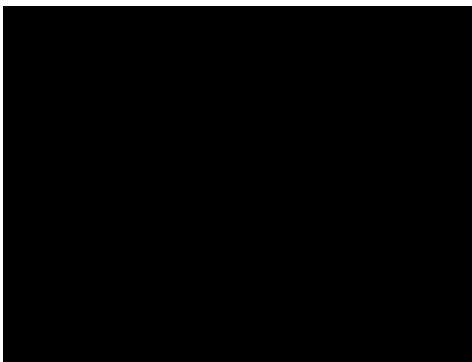
ButtonBar

Button

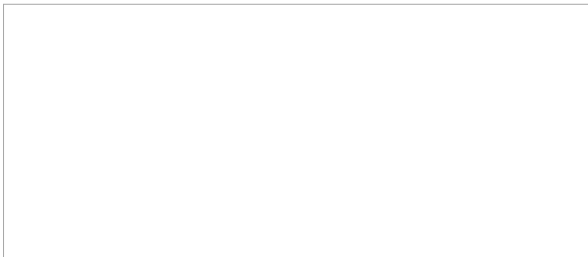
RadioButton

ButtonGroup

Button



18: Dawn of War II -



19: RadioButton

RadioButton (gfx.controls.RadioButton) 1
 RadioButton 1
 RadioButton

CLIK RadioButton CheckBox
 RadioButton
 ButtonGroup ()
 () RadioButton toggle
 ButtonGroup

RadioButton
 RadioButton
 RadioButton
 ButtonGroup
 RadioButton
 Button
 RadioButton

CLIK RadioButton MovieClip
 :
 • ***nrnc*** : () TextField
 • ***focusIndicator*** : () MovieClip
 MovieClip
 "hide" 2 MovieClip "show"

•

•

Getting Started with CLIK Buttons

toggle disableFocus

27

	textField		Button
	textField	Button	
	textField	/	
selected	true	RadioButton	(
Data	RadioButton		)
group	ButtonGroup	RadioButton	RadioButton
		ButtonGroup	RadioButton
		ButtonGroup	RadioButton _root
			_root
	RadioButton	1	

1 Object

- *rs* :
- *n la n4*

RadioButton

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
select	selected	selected
	<i>selected</i> : Button	selected
stateChange	<i>state</i> : Button	
	"up" "over" "down"	
	Getting Started with CLIK Buttons	
rollOver	<i>mouseIndex</i> :	

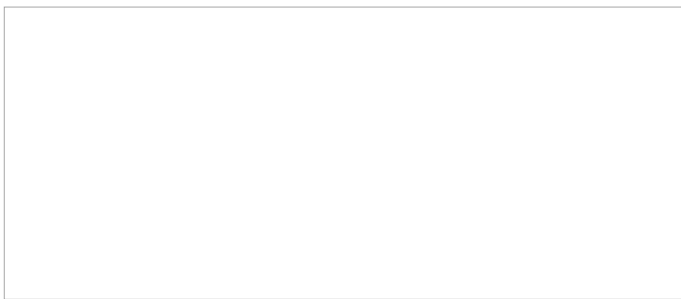
	<code>(</code>	<code>0</code>	<code>3</code>
rollOut	<code>)</code>		
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	
press	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	
doubleClick	<code>true</code>	<code>doubleClickEnabled</code>	
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	
click	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	
dragOver	<code>(</code>	<code>)</code>	
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	
dragOut	<code>(</code>	<code>)</code>	
	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	
releaseOutside	<code>mouseIndex :</code>	<code>(</code>	<code>0</code>
	<code>)</code>	<code>3</code>	

RadioButton :

```
myRadio.addEventListener("select", this, "onRadioToggle");
function onRadioToggle(event:Object) {
    //
}
```

RadioButton :

```
import gfx.controls.ButtonGroup;
var myGroup:ButtonGroup = new ButtonGroup("groupName");
radio1.group = myGroup;
radio2.group = myGroup;
someMovie.radio1.group = myGroup;
someMovie.radio2.group = myGroup;
```



21: ButtonGroup

CLIK ButtonGroup (gfx.controls.ButtonGroup)

1

ButtonGroup

CLIK

Button (CheckBox RadioButton)
ButtonGroup

ButtonGroup

RadioButton

CLIK ButtonGroup

MovieClip

ButtonGroup

ButtonGroup

1 Object

- *item* :
- *target* :

ButtonGroup

change	• <i>item</i> : • <i>data</i> :	CLIK Button AS2 Object
itemClick	• <i>item</i> :	CLIK Button

ButtonGroup :

```
myGroup.addListener("change", this, "onNewSelection");
function onNewSelection(event:Object) {
    if (event.item == myRadio) {
        //
    }
}
```

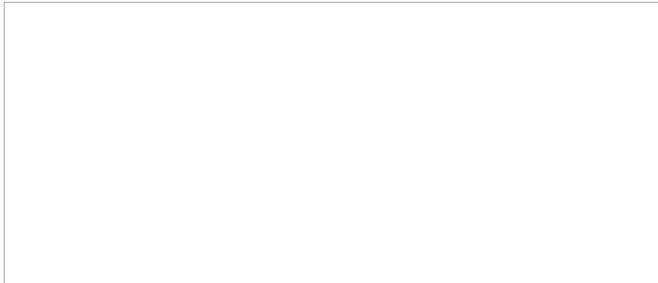
RadioButton :

```
var selectedRadio:Button = group.selectedButton;
```

RadioButton
:

ButtonGroup

```
radioBtn1.group.addEventListener("itemClick", this, "onClick");
```



22: ButtonBar

ButtonBar (gfx.controls.ButtonBar) ButtonGroup
dataProvider
(dataProvider
) ButtonBar
dataProvider
UI
Button
:

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

ButtonBar ButtonGroup
ButtonBar Button

CLIK ButtonBar MovieClip

CLIK ButtonBar

Button

ButtonBar

(Flash IDE

)
ButtonBar

Button

visible	false	ButtonBar	
disabled	true	ButtonBar	
itemRenderer	Button ButtonBar	ID	
direction	Button horizontal vertical	Button	
spacing	Button (<i>direction</i>)		
autoSize	true	Button	
buttonWidth	true	Button	autoSize

1 Object

- *type* :
- *target* :

ButtonBar

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	• <i>index</i> : ButtonBar 1	0

	• <i>item</i> : Button CLIK Button • <i>item</i> : dataProvider AS2 Object • <i>data</i> : dataProvider AS2 Object
itemClick	• <i>index</i> : Button ButtonBar 0 1 • <i>item</i> : dataProvider AS2 Object • <i>data</i> : dataProvider AS2 Object • <i>mouseIndex</i> : (0 3)

ButtonBar Button
:

```
myBar.addEventListener("itemClick", this, "onItemClick");
function onItemClick(event:Object) {
    processData(event.data);
    //
}
```

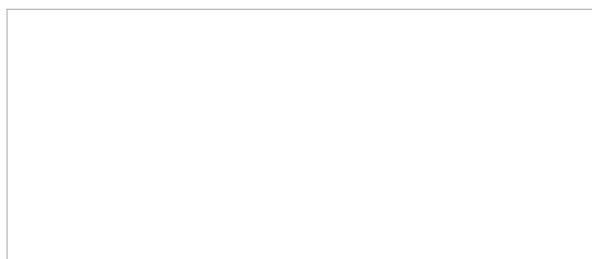


```

ScrollIndicator ScrollBar Slider
ScrollIndicator
)
ScrollIndicator 4 ScrollBar
Slider ScrollBar
Slider ScrollBar

```

23: *Mercenaries 2* -



24: ScrollIndicator

```

CLIK ScrollIndicator (gfx.controls.ScrollIndicator) textField
ScrollIndicator

```

```

        textField
        TextArea
        ScrollIndicator
        ID
        ScrollBar (
        scrollbar
    )

```

ScrollIndicator

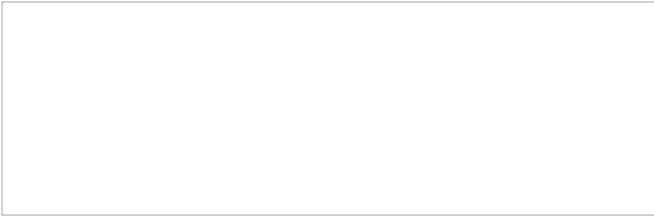
CLIK ScrollIndicator MovieClip

:

- *thumb* : CLIK Button
- *track* : MovieClip

ScrollIndicator thumb

track Button



25: ScrollIndicator

ScrollIndicator :

scrollTarget	TextArea	textField	textField
	ScrollIndicator		
visible	false		
disabled	true		
offsetTop	thumb		

offsetBottom	thumb
--------------	-------

1 Object

- *type* :
- *target* :

ScrollIndicator

show	visible	true
hide	visible	false
scroll	• <i>position</i> :	

:

```
mySI.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

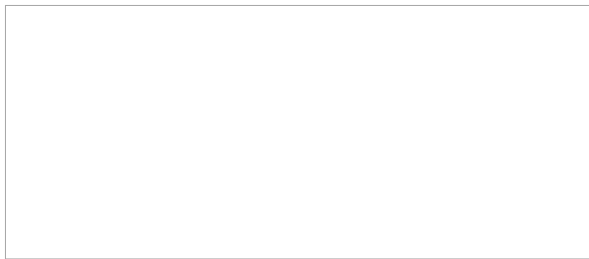
:

```
scrollInd.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollInd.positon = currPos;
```

_rotation
ScrollIndicator

:

```
scrollInd.direction = "horizontal";
```



26: ScrollBar

CLIK ScrollBar (gfx.controls.ScrollBar)

ScrollIndicator

ScrollBar

ScrollBar

ScrollBar

2

1

1

trackMode inspectable
) trackMode

Shift (

CLIK ScrollBar

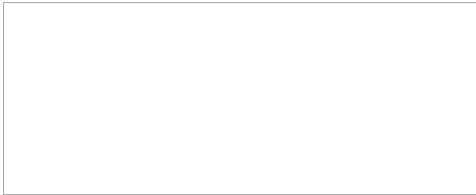
MovieClip

:

- *upArrow* : CLIK Button
- *downArrow* : CLIK Button
- *thumb* : CLIK Button
- *track* : CLIK Button

Button

Button



27: ScrollBar

ScrollIndicator ScrollBar
thumb up down track Button

ScrollBar ScrollIndicator 1 :

scrollTarget	TextArea	textField	textField
ScrollIndicator			
trackMode	scrollPage scrollToCursor		
visible	false		
disabled	true		
offsetTop	thumb		
offsetBottom	thumb		

1 Object

- *type* :
- *target* :

ScrollBar

show	visible	true
hide	visible	false

scroll

- *position:*

:

```
mySB.addEventListener("scroll", this, "onListScroll");
function onListScroll(event:Object) {
    trace("scroll position: " + event.position);
    //
}
```

:

```
scrollBar.setScrollProperties(pageSize, minPos, maxPos, pageScrollSz);
scrollBar.positon = currPos;
```

_rotation
ScrollBar

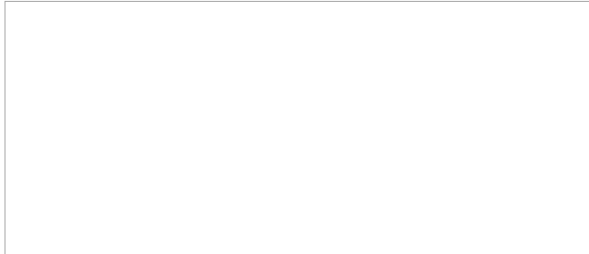
:

```
scrollBar.direction = "horizontal";
```

scrollPage

1 :

```
scrollBar.trackScrollPageSize = pageSize;
```



28: Slider

Slider (gfx.controls.Slider)

Slider

Slider

Home

End

CLIK Slider

MovieClip

:

- *thumb* : CLIK Button
- *track* : CLIK Button

ScrollIndicator ScrollBar Slider
thumb track Button



図 29: Slider のタイムライン

Slider :

visible	false			
disabled	true			
value	Slider			
minimum	Slider			
maximum	Slider			
snapping	true	snapInterval		
snapInterval	snapInterval false		snapping	
liveDragging	true	Slider false	Slider	change
	change			
offsetLeft	thumb			
offsetRight	thumb			

1 Object

- *type* :
- *target* :

Slider

Show	visible	true
Hide	visible	false
focusIn		
focusOut		
change	Slider	

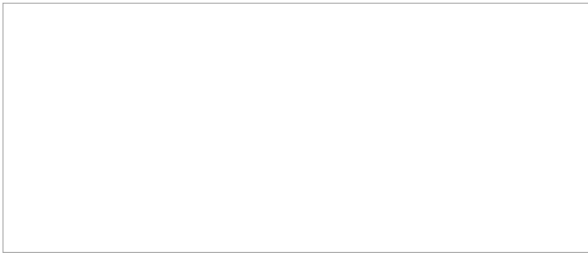
```
Slider :
mySlider.addListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("slider value: " + event.target.value);
    //
}
```

CLIK	NumericStepper	OptionStepper	ScrollingList	TileList
DropDownMenu		NumericStepper		
	DataProvider			
ListItemRenderer			ScrollingList	TileList
NumericStepper	OptionStepper	1		
ScrollingList	TileList			2
	ScrollIndicator	ScrollBar		
	DropDownMenu		1	
		ScrollingList		TileList

30: *Mercenaries 2* -

31: *Halo Wars* -

32: *Mass Effect* -



33: NumericStepper

NumericStepper (gfx.controls.NumericStepper) 1

NumericStepper 2

Home End

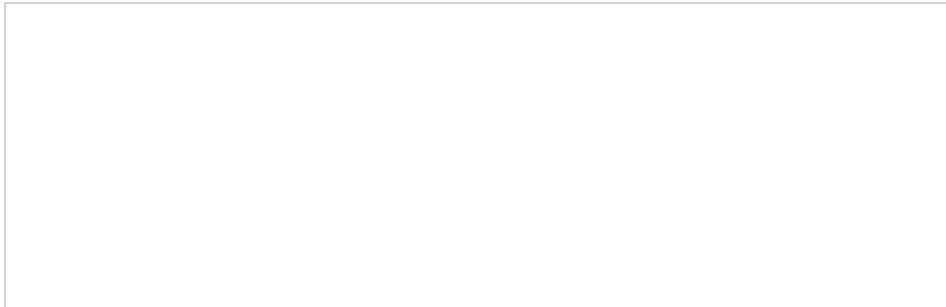
NumericStepper MovieClip :

- *textField* : TextField
- *nextBtn* : CLIK Button
- *prevBtn* : CLIK Button

NumericStepper : focused disabled 3

- *default*
- *focused* textField

- *disabled*



34: NumericStepper

NumericStepper

:

MovieClip

visible	false
disabled	true
value	NumericStepper
minimum	NumericStepper
maximum	NumericStepper

1 Object

- *type :*
- *targe :*

NumericStepper

show	visible	true
hide	visible	false
change	NumericStepper	
stateChange	NumericStepper	focused disabled
	• <i>state :</i>	"default" "focused" "disabled"

NumericStepper

:

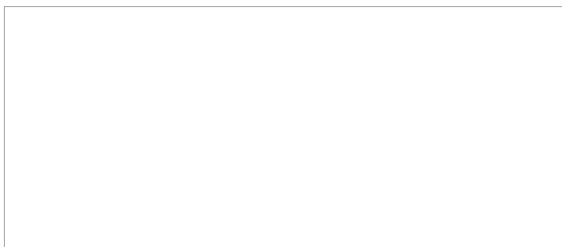
```
myNS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("ns value: " + event.target.value);
    //
}
```

:

```
ns.stepSize = 0.5;
```

:

```
ns.labelFunction = function(value:Number) {
    switch(value) {
        case 1:
            return value + "st";
        default:
            return value + "th";
    }
}
```



35:

OptionStepper

NumericStepper

OptionStepper (gfx.controls.OptionStepper) 1

dataProvider

OptionStepper selectedIndex

dataProvider dataProvider 0-
 dataProvider :

```
optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

NumericStepper OptionStepper 2

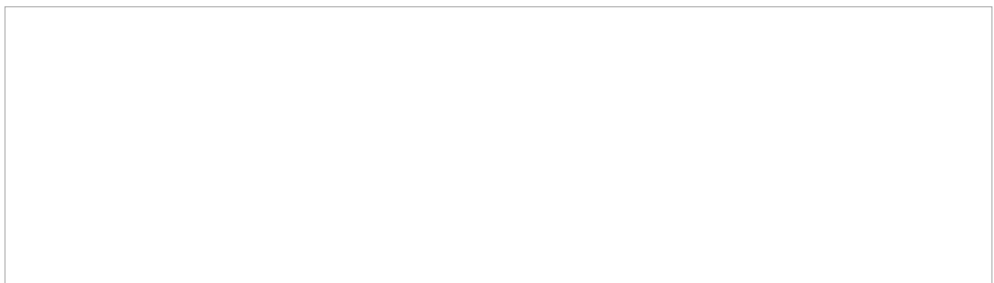
Home End dataProvider

NumericStepper MovieClip :

- *textField* : TextField
- *nextBtn* : CLIK Button dataProvider
- *prevBtn* : CLIK Button dataProvider

OptionStepper	:	focused	disabled	3
---------------	---	---------	----------	---

- *default*
- *focused* textField
- *disabled*



36: OptionStepper

OptionStepper	MovieClip
:	

visible	false
disabled	true

1 Object

- *type* :
- *target* :

OptionStepper

show	visible	true
------	---------	------

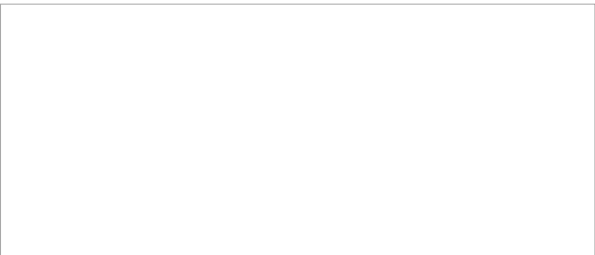
hide	visible		false
change	OptionStepper		
stateChange	OptionStepper	focused	disabled
	state : "default" "focused" "disabled"		

OptionStepper :

```

myOS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("os value: " + event.target.selectedItem);
    //
}

```



37: ListItemRenderer

ListItemRenderer (gfx.controls.ListItemRenderer) CLIK Button

ScrollingList TileList DropdownMenu

ListItemRenderer Button
Button

ListItemRenderer

ListItemRenderer

CLIK ListItemRenderer

MovieClip

-
-

- *textField* : () TextField
- *focusIndicator* : () MovieClip
 MovieClip MovieClip "show"
 "hide" 2

ListItemRenderer

selected

•

•

- *up*
-
- *over*
- *down*
- *disabled*
- *selected_up*
-
- *selected_over*
- *selected_down*
- *selected_disabled*

38: ListItemRenderer

ListItemRenderer

Button
ListItemRenderer

Getting Started with CLIK Buttons

		)	0	3
doubleClick			<i>doubleClickEnabled</i>	
		true		

```
scrollingList.dataProvider = ["item1", "item2", "item3", "item4"];
```

ScrollingList

itemRenderer

ListItemRenderer

ListItemRenderer

FLA

[

]

ScrollingList

ScrollBar

1

ScrollingList

wrapping

ScrollingList
"normal"

wrapping

wrapping
wrapping

"wrap"
"stick"

Page Up

Page Down

End

Home

ScrollingList

ScrollBar

ScrollBar

ScrollBar

D

CLIK

UI

ScrollingList

ScrollingList

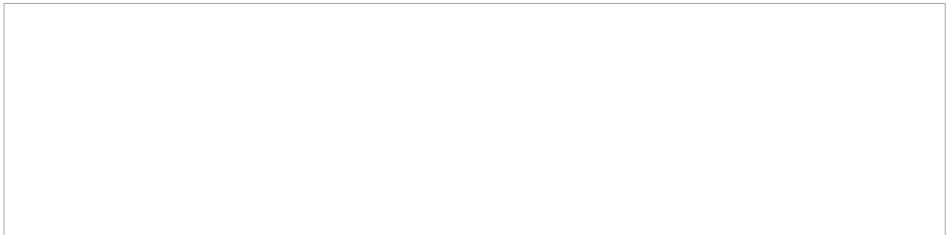
ScrollingList

:

focused disabled

3

- *default*
- *focused*
- *disabled*



40: ScrollingList

ScrollingList

:

MovieClip

visible	false
disabled	true
itemRenderer	ListItemRenderer
renderInstanceName	ScrollingList
scrollBar	ScrollBar
	ScrollingList

ScrollingList		
margin	<i>rendererInstanceName</i>	
rowHeight	<i>rendererInstanceName</i>	
paddingTop	rendererInstanceName	
padding Bottom	rendererInstanceName	
paddingLeft	rendererInstanceName	
paddingRight	rendererInstanceName	
thumbOffsetTop		
thumbOffsetBottom		
thumbSizeFactor	1.0	1.0

1 Object

- *type* :
- *target* :

ScrollingList

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	<i>index</i> : 0 1	
itemPress	<i>renderer</i> : CLIK Button <i>item</i> : dataProvider AS2 Object <i>index</i> : dataProvider 0 1 <i>mouseIndex</i> : (0 3)	
itemClick	<i>renderer</i> : CLIK Button <i>item</i> : dataProvider AS2 Object <i>index</i> : dataProvider 0 1 <i>mouseIndex</i> : (0 3)	
itemDoubleClick	<i>renderer</i> : CLIK Button <i>item</i> : dataProvider AS2 Object <i>index</i> : dataProvider 0 1 <i>mouseIndex</i> : (	

	)	0	3
itemRollOver			
• <i>renderer</i> :	Button		CLIK
• <i>item</i> :	dataProvider	AS2	Object
• <i>index</i> :	dataProvider	0	1
• <i>mouseIndex</i> :	(		
	)	0	3
itemRollOut			
• <i>renderer</i> :			CLIK Button
• <i>item</i> :	dataProvider	AS2	Object
• <i>index</i> :	dataProvider	0	1
• <i>mouseIndex</i> :	(		
	)	0	3

:

```
myList.addEventListener("itemClick", this, "onItemClicked");
function onItemClicked(event:Object) {
    trace("list item was clicked: " + event.renderer);
    //
}
```

1 :

```
//
// ScrollingList 'label' :
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

// label
//
```

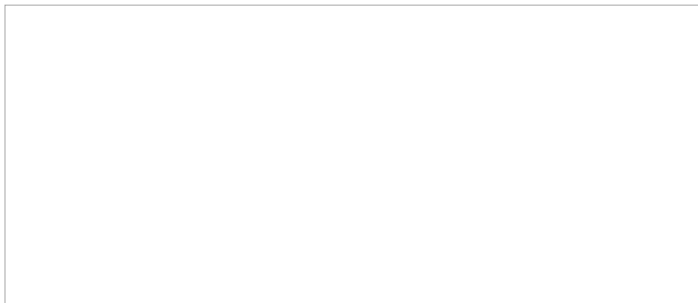


```

//      :
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

//
//      labelFunction
//      :
list.labelFunction = function(itemObj:Object):String {
//
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```



41: TileList

TileList (gfx.controls.TileList) ScrollingList

ScrollBar ScrollIndicator

TileList ScrollingList TileList

dataProvider 4 dataProvider

:

```
tileList.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

TileList ListItemRenderer

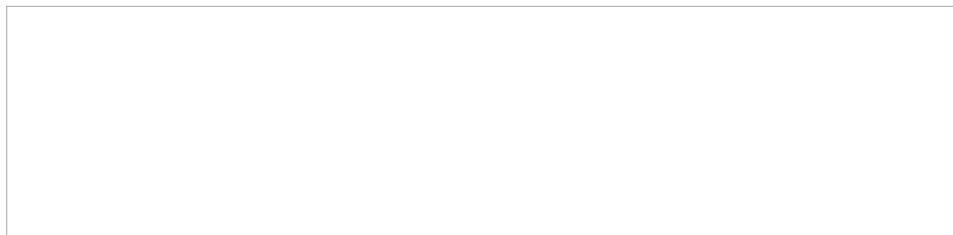
ListItemRenderer FLA []



TileList
TileList

TileList : focused disabled 3

- *default*
- *focused*
- *disabled*



42: TileList

TileList
:

MovieClip

1 Object

- *type* :
- *target* :

TileList

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	• <i>index</i> : 0 1	
itemPress	• <i>renderer</i> : CLIK Button • <i>item</i> : dataProvider AS2 Object • <i>index</i> : dataProvider 0 1 • <i>mouseIndex</i> : (0 3)	
itemClick	• <i>renderer</i> : CLIK Button • <i>item</i> : dataProvider AS2 Object • <i>index</i> : dataProvider 0 1 • <i>mouseIndex</i> : (0 3)	

itemDoubleClick

- *renderer* : CLIK
Button
- *item* :
dataProvider AS2 Object
- *index* : dataProvider 0 1
- *mouseIndex* :
(0 3)

itemRollOver

- *renderer* : CLIK
Button
- *item* :
dataProvider
- *index* : dataProvider 0 1
- *mouseIndex* :
(0 3)

itemRollOut

- *renderer* : CLIK Button
- *item* :
dataProvider AS2 Object
- *index* : dataProvider 0 1
- *mouseIndex* :
(0 3)

TileList

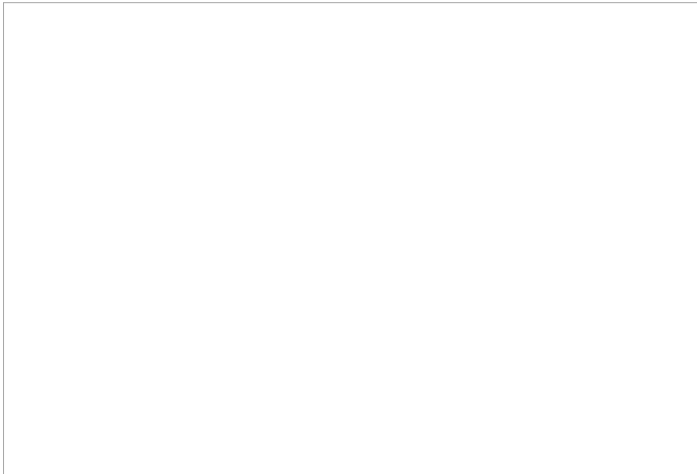
:

```
myList.addEventListener("focusIn", this, "onListFocused");
function onListFocused(event:Object) {
    trace("tile list was focused!");
    //
}
```

1

:

```
//  
// TileList      'label'      :  
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];  
  
//      label  
//  
//      :  
list.labelField = "name";  
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];  
  
//      labelFunction  
//      :  
list.labelFunction = function(itemObj:Object):String {  
//  
}  
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```



43: DropdownMenu

DropdownMenu (gfx.controls.DropdownMenu)

DropdownMenu

ScrollBar

ScrollIndicator

ScrollingList

TileList

dataProvider

DropdownMenu
dataProvider

dataProvider
:

```
dropdownMenu.dataProvider = ["item1", "item2", "item3", "item4"];
```

DropdownMenu

ScrollingList

dropdown

ScrollingList ListItemRenderer

FLA

[

]

DropdownMenu
ScrollBar

ScrollIndicator

DropdownMenu

DropDownMenu

Enter

ScrollingList TileList ScrollBar

DropDownMenu

DropDownMenu

DropDownMenu

DropDownMenu

Button

MovieClip

:

- *textField* : () TextField
- *focusIndicator* : () MovieClip
- "hide" 2 MovieClip "show"

DropDownMenu

ToggleButton

CheckBox

:

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

44: DropdownMenu

DropdownMenu

Button
DropdownMenu

[Getting Started with CLIK Buttons](#)

DropdownMenu

:

visible	false
disabled	true
dropdown	DropdownMenu (ScrollingList TileList)
dropdownWidth	DropdownMenu -1
itemRenderer	
scrollBar	
margin	

thumbSizeFactor	1.0		
offsetX			
offsetY			
extent	offsetX dropdownWidth	-1	
direction		"up"	"down"

1 Object

- *type* :
- *target* :

DropdownMenu Button *change*

show	visible	true
hide	visible	false
focusIn		
focusOut		
change	• <i>index</i> : 0 1 • <i>data</i> : AS2 Object	
select	selected	
	• <i>selected</i> : Button selected	
stateChange	• <i>state</i> : Button "up" "over" "down"	

[Getting Started with CLIK Buttons](#)

rollOver	• <i>mouseIndex</i> :
rollOut	• <i>mouseIndex</i> :
press	• <i>mouseIndex</i> :
doubleClick	• <i>mouseIndex</i> :
click	• <i>mouseIndex</i> :
dragOver	• <i>mouseIndex</i> :
dragOut	• <i>mouseIndex</i> :
releaseOutside	• <i>mouseIndex</i> :

:

```
dropdown.addEventListener("click", this, "onClick");
function onClick(e:Object) {
    if (e.target.isOpen) {
```

```

    //
  }
}

```

DropDownMenu :

```

// ID ;
// ID /
// fla
// DropdownMenu ScrollingList ScrollBar
attachMovie("DropDownMenu", "dd", this.getNextHighestDepth(),
            {dropdown: "ScrollingList"});
dd.dataProvider = ["one", "two", "three", "four", "five", "six"];

//
//
//
//
//
onEnterFrame = function() {
    dd.dropdown.scrollBar = "ScrollBar";
    onEnterFrame = null;
}

```

1 :

```

//
// DropdownMenu 'label' :
list.dataProvider = [{label: "one", data:1}, {label: "two", data:2}];

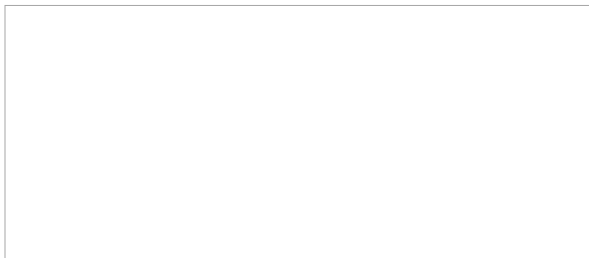
// label
//
// :
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two", data:2}];

//
// labelFunction :
list.labelFunction = function(itemObj:Object):String {
    //
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

Scaleform CLIK
ProgressBar 2
StatusIndicator
()
StatusIndicator
ProgressBar

45: *Mercenaries 2* - FACTIONS



46: StatusIndicator

StatusIndicator (gfx.controls.StatusIndicator) StatusIndicator

StatusIndicator

StatusIndicator

visible	false
disabled	true
Value	
minimum	
maximum	

- *type* :
- *target* :

Show	visible	true
Hide	visible	false

ProgressBar (gfx.controls.ProgressBar)
StatusIndicator

ProgressBar

2

```
ProgressBar
gotoAndStop()
```

value

73

disabled	true		
Target	bytesLoaded	bytesTotal	ProgressBar
Mode	ProgressBar	"manual"	
	setProgress		
	"polled"	bytesLoaded	bytesTotal
		"event"	
	"progress"	bytesLoaded	bytesTotal

1 Object

- *type* :
- *target* :

ProgressBar

Show	visible	true
Hide	visible	false
progress	ProgressBar	
complete	ProgressBar	

:

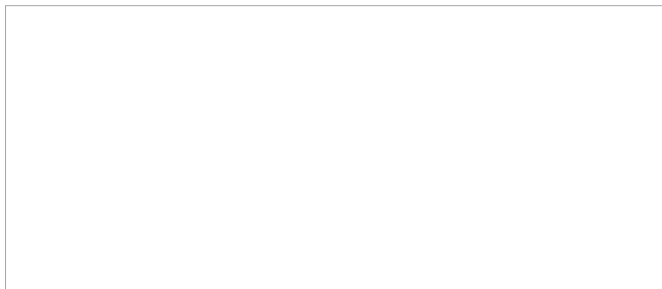
```

myProgress.addEventListener("progress", this, "onProgress");
myProgress.addEventListener("complete", this, "onProgress");
function onProgress(event:Object) {
    if (event.type == "progress") {
        //
    } else
    {
        trace("Loading complete!");
    }
}

```


Scaleform CLIK
 UI
 Dialog UILoader ViewStack Dialog
 UILoader
 ViewStack

48: *Halo Wars* -



49: Dialog

CLIK Dialog (gfx.controls.Dialog)

MovieClip
 1 MovieClip () Dialog.show()

Dialog

CLIK Dialog

MovieClip

:

- closeBtn* : () CLIK Button
- cancelBtn* : () CLIK Button
- submitBtn* : () CLIK Button
- dragBar* : () MovieClip

OK

Dialog

MovieClip

Dialog

MovieClip

:

visible	false
disabled	true

- *type* :
- *target* :

Dialog

Show	visible	true
Hide	visible	false
Submit	Dialog • <i>data</i> : Dialog getSubmitData	Dialog true () AS2 Object
Close	Dialog	

:

```
myDialog.addEventListener("submit", this, "onLoginSubmit");
function onLoginSubmit(event:Object) {
    trace("Received data from dialog: " + event.data);
}
```

CLIK UILoader (gfx.controls.UILoader)
UILoader

SWF/GFX

Scaleform

UILoader

SWF/GFX UILoader

CLIK UILoader

MovieClip

:

- *bg* : () MovieClip UILoader

UILoader

SWF/GFX

UILoader

UILoader

MovieClip

:

visible	false		
autoSize	true	UILoader	
maintainAspectRatio	true		UILoader false
		UILoader	
Source		SWF/GFX	
timeout			
	UILoader	'ioError'	

1 Object

- *type* :
- *target* :

UILoader

Show	visible	true
Hide	visible	false
progress	a)	b)

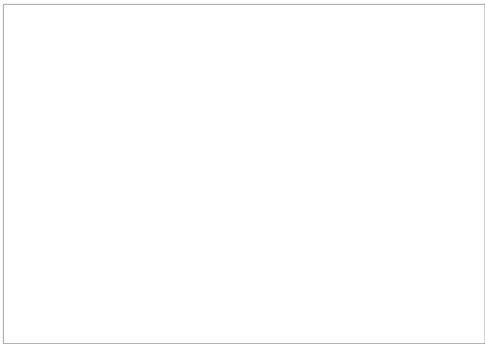
Loaded	0	100
complete		
ioError	source	

:

```

myUILoader.addEventListener("ioError", this, "onLoadingError");
function onLoadingError(event:Object) {
    displayErrorMessage("Could not load" + event.target.source);
}

```



50: ViewStack

CLIK ViewStack (gfx.controls.ViewStack)
1

Accordion

RadioButton

demos

)

TabBox
(2
ViewStack

ViewStack

ViewStack

CLIK ViewStack

MovieClip

ViewStack

ViewStack

ViewStack

:

MovieClip

visible	false
Cache	true
ViewStack targetGroup ()	
targetGroup	ButtonGroup 'change'
ID Flash IDE [	
RadioButton ID	
]	

ViewStack

Scaleform CLIK

Scaleform CLIK

CLIK

Flash

:

1. Flash []
2. [] []
3. Flash []
4. [] []
5. 1 px
6. [OK]

51: []

: 2 2 SWF

52:

Flash

Scaleform

Flash

PNG Photoshop®

1

()

Flash

1

()

2

2

:

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

Flash

Scaleform CLIK

Flash

2

Button FLA () CLIK

FLA

:

2 *CLIK_Components fla*

3 []

(Button)

FLA [

[]

4 [] FLA ['Button'

5 [] FLA []

6 []

3

[]

[]

[]

()

:

1. []

[]

2. []
(gfx.controlsButton)

[]

Ctrl+C

3. []

4.

5. [] [ActionScript []]

6. []

Ctrl+V

7. [OK]

Ctrl+V

$$(\quad)$$
$$([\quad])$$

Scaleform CLIK

1

CLIK

CLIK
FLA []

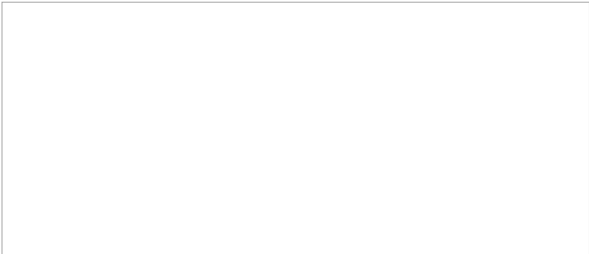
Flash

:

- Flash
- Flash
- Photoshop Illustrator®

Flash

[Getting Started with CLIK](#)



55: StatusIndicator

StatusIndicator

Button

StatusIndicator

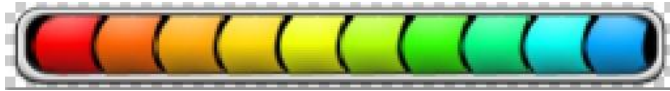
indicator track 2 *track*

indicator

1 Photoshop

StatusIndicator

1
StatusIndicator



56: StatusIndicator PSD

57: StatusIndicator PSD Photoshop

1. StatusIndicator Photoshop
2. PSD
3. Flash [] []->[]
4. StatusIndicator PSD
5. [] [] [Flash]
6. [OK]
7. PSD Flash

10

1

(PSD 10)

10

Flash 10

1

[1 10]

1
- 8.
- 9.
10. [1] 6

11. [1] 11 11
 12. [2] []
 13. [1] 11 Ctrl+X Ctrl+Shift+V
 14. 10 [1] (1) 5
 11 [3] 16 2] [4]
 21 10
 PSD 51
 15. (2-10)
 16. 5 []
 55
 17. *indicator*
 18. *track* ° ° *track*
 19. [1] 'indicator'
 20. *indicator* *track* *actions*
track *indicator*

58: ()

21. StatusIndicator
 22. 1 10
 23.
 24.

59: StatusIndicator

Scaleform CLIK

Scaleform () SWF
Scaleform
Flash UI

Flash Player Scaleform
Scaleform Player

Scaleform
Slate Mobile Slate Mobile textField
(CJK)
ASCII
Slate Mobile CJK

Scaleform textField
) Scaleform (3.4.4
Scaleform Scaleform
UI

textField
) ([
[OK]
FLA textField
FLA
textField
(
textField
] ([]->[
FLA

```

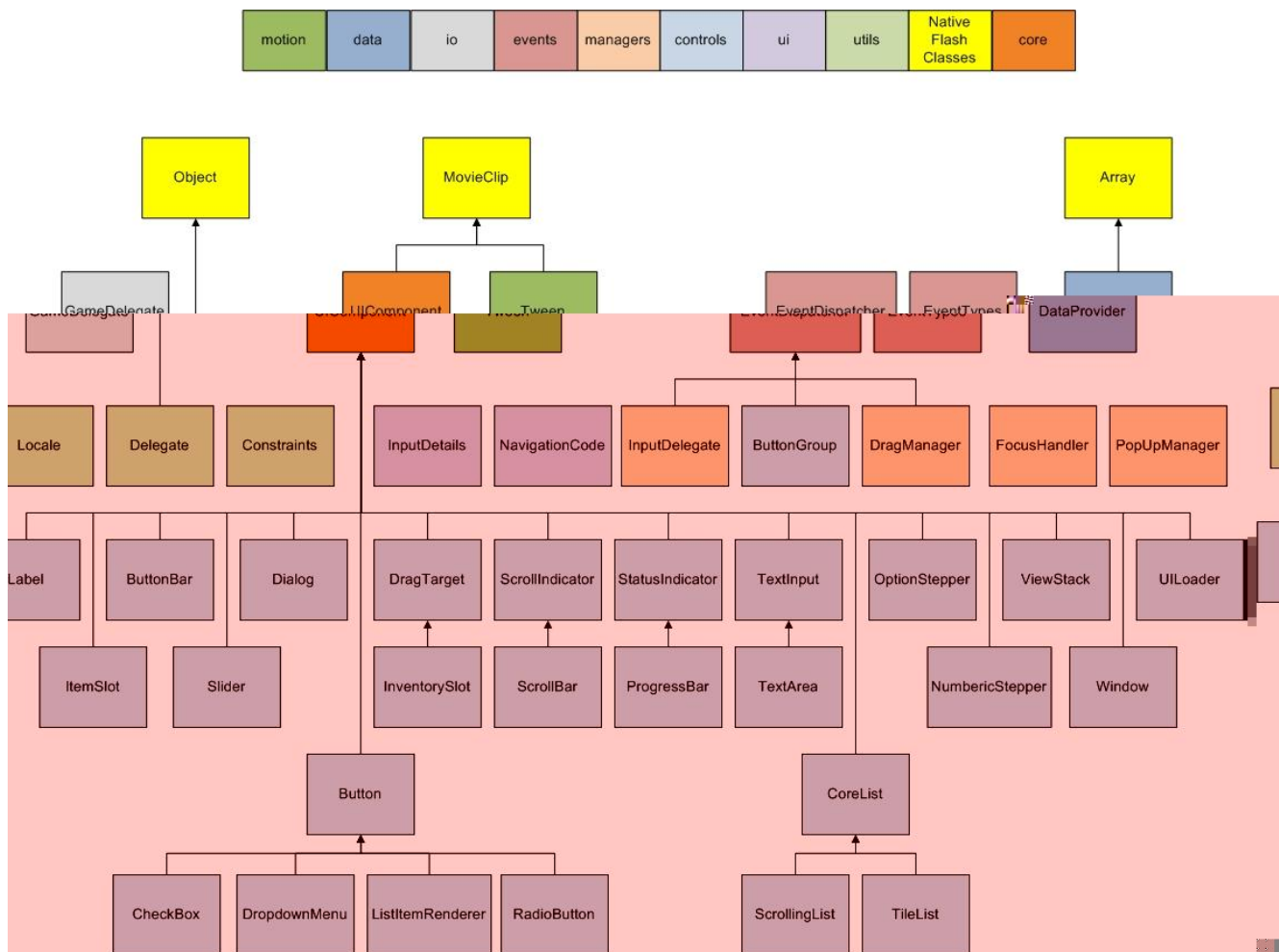
Scaleform
/
SWF
Scaleform
SWF
Scaleform
textField
textField
gfxfontlib fla
Scaleform
textField
) Scaleform (
fontconfig txt
textField
fontconfig txt
Scaleform
textField '$TITLE'
$TITLE=Scaleform
textField 'Scaleform'
Scaleform
Scaleform

```

[Font Overview](#)

CLIK

Scaleform



60: Scaleform CLIK

C i i h hn

Scaleform CLIK

UIComponent (gfx.core.UIComponent)
UIComponent CLIK

UIComponent Scaleform

UIComponent Flash 8 MovieClip UIComponent
Flash MovieClip / :

- *disabled;*
 - *visible;*
 - *focused;*
 - *width;*
 - *height;*
 - *displayFocus :* focused
- true

```
UIComponent
:
```

- *configUI* :
- *draw* :

- *b ha i om4*

- *mli* *b* :

UIComponent Mix-in EventDispatcher

onLoad() :

- MovieClip

- configUI()
- validateNow()
- draw()

Scaleform CLIK

3

CheckBox Button ListItemRenderer CLIK RadioButton
 ScrollBar
 (gfx.controls.Button) CLIK Button
 Button

:

- *up*
- *over*
- *down*
- *disabled*

"selected_"

CLIK Button

selected

:

- *up*
- *over*
- *down*
- *disabled*

- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

: CLIK

[Getting Started with CLIK](#)

Buttons

CLIK Button getStatePrefixes()

:

```
private function getStatePrefixes():Array {
    return (_selected) ? ["selected_", ""] : [""];
}
```

CLIK
getStatePrefixes() "selected_"
selected

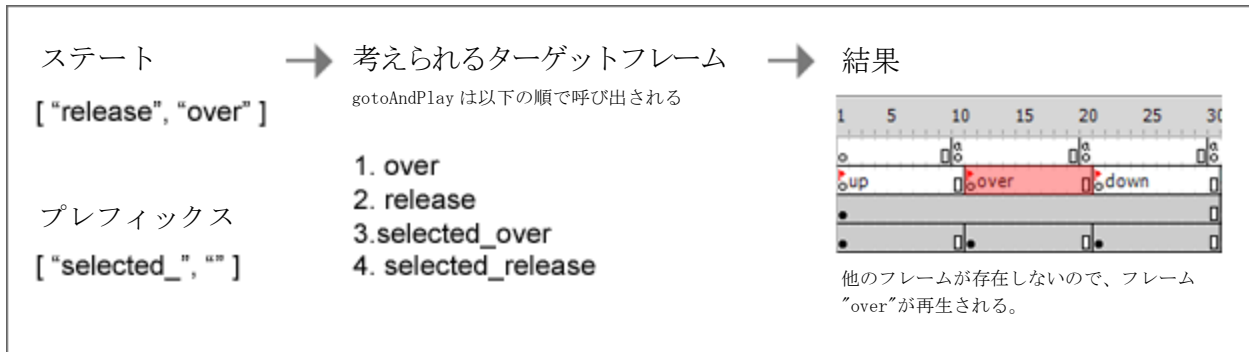
Button stateMap
CLIK Button

:

```
private var stateMap:Object = {
    up: ["up"],
    over: ["over"],
    down: ["down"],
    release: ["release", "over"],
    out: ["out", "up"],
    disabled: ["disabled"],
    selecting: ["selecting", "over"],
    kb_selecting: ["kb_selecting", "up"],
    kb_release: ["kb_release", "out", "up"],
    kb_down: ["kb_down", "down"]
}
```

getStatePrefixes()

:



61:

CLIK

ScrollingList OptionStepper Slider TextArea

:

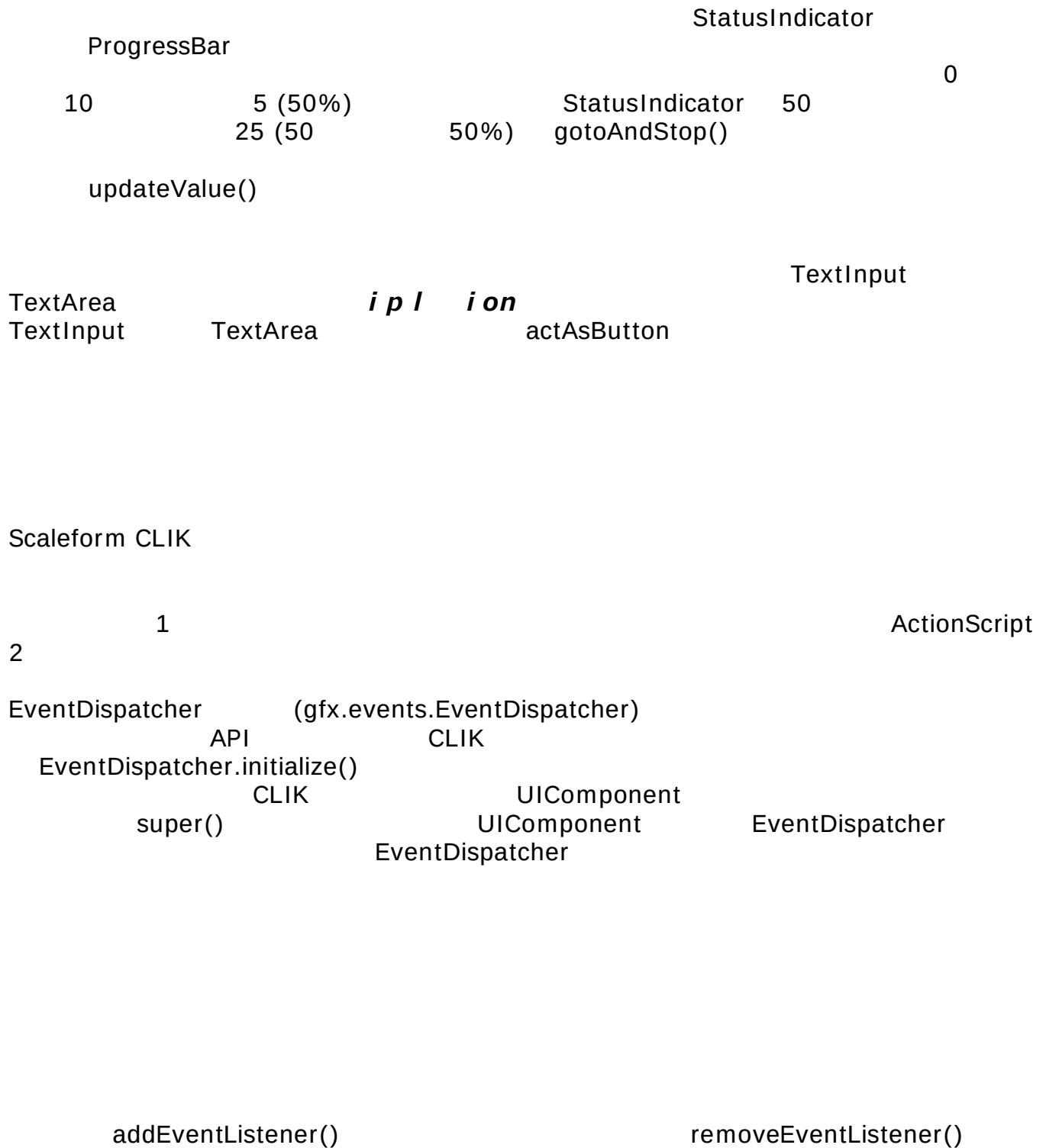
- *default*
- *focused*
- *disabled*

Label

:

- *default*

- *disabled*



1

callBack
String

callBack

EventTypes (gfx.events.EventTypes)

("show"

EventTypes.SHOW)

```
buttonInstance.addEventListener("itemClick", this, "callBack");  
function callBack(eventObj:Object):Void {  
    buttonInstance.removeEventListener("itemClick", this, "callBack");  
}
```

buttonInstance "itemClick"
"callback"

callback

event Object

CLIK

()

dispatchEvent()

CLIK
1

target

```
dispatchEvent({type:"itemClick", item:selectedItem});
```

Scaleform CLIK

Scaleform Player

()
Selection.setFocus(instance)
(gfx.managers.FocusHandler) 1
FocusHandler

ActionScript
FocusHandler

```

        CLIK
        Player
        (
            Tab
            focused
            true
        )
    :

```

```

Selection.setFocus(firstComponentOrMovieClip);

```

```

        MovieClip (onRelease onRollOver ) Scaleform Flash
Player

```

```

MovieClip focusEnabled tabEnabled

```

```

ScrollBar
focusTarget

```

```

        focusTarget
        FocusHandler
        focusTarget

```

```

        Player

```

```

displayFocus
        true

```

```

Slider

```

```

        Slider
        UIComponent.changeFocus()
        focused

```

```

function changeFocus():Void {
    track.displayFocus = _focused;
}

```

```

        tabEnabled
        false

```

```

background.tabEnabled = false;

```


ScrollingList OptionStepper ButtonBar

Flash Scaleform Selection

:

1. (OptionStepper
2.) tabEnabled = false
 focusTarget
3. focusEnabled = true
 changeFocus displayFocus
 true

Scaleform CLIK

Flash 8 MovieClip
MovieClip

CLIK

InputDelegate
(gfx.managers.InputDelegate) InputDelegate

InputDetails (gfx.ui.InputDetails)
InputDelegate.readInput()

InputDetails
InputDelegate

InputDetails :

-
-
-

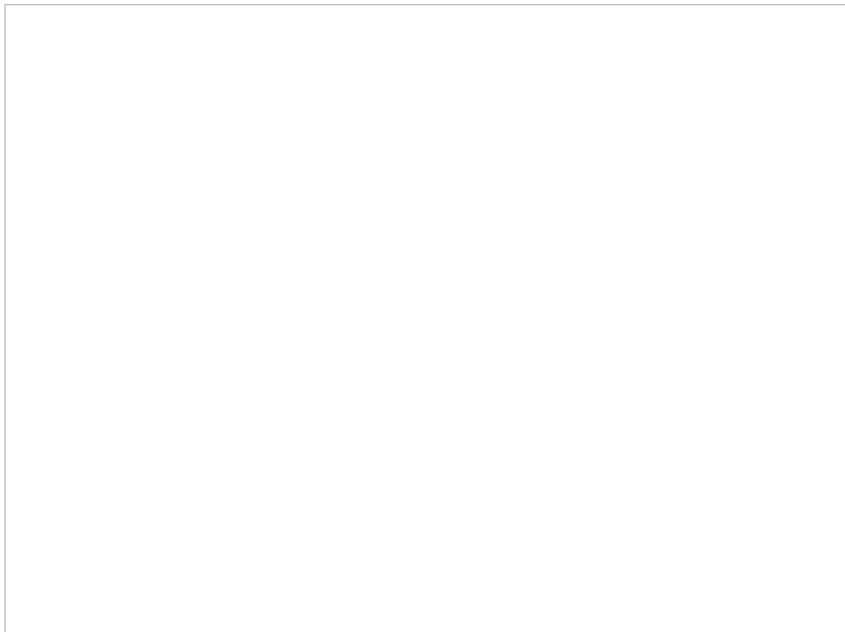
key up key down
 InputDetails
 “keyUp” “keyDown”
 •
 NavigationCode navEquivalent ()
 (gfx.ui.NavigationCode)

FocusHandler InputDelegate

handleInput()

pathToFocus

InputDetails



62: handleInput()

FocusHandler handleInput()

false null navEquivalent
 Scaleform Player

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
```

```

        if (details.navEquivalent == "left")
        {
            //      NavigationCode.LEFT
            doSomething();
            return true;
        }
        return false;
    }
}

```

true pathToFocus
 false
 handleInput

```

function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var nextItem:MovieClip = pathToFocus.shift();
    var handled:Boolean = nextItem.handleInput(details, pathToFocus);
    if (handled) { return true; }
    //
    return false; //
}

```

pathToFocus DropdownMenu handleInput

```

function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var handled:Boolean = dropdown.handleInput(details);
    if (handled) { return true; }
    //
    return false; //
}

```

InputDelegate.instance

FocusHandler

```

InputDelegate.instance.addEventListener("input", this, "handleInput");
function handleInput(event:Object):Void {
    var details:InputDetails = event.details;
    if (details.value == Key.TAB) { doSomething(); }
}

```

InputDelegate

InputDelegate

W A S D

Wii

1

SWF

CLIK

mouseIndex

```
myButton.addEventListener("click", this, "onCursorClick");  
function onCursorClick(event:Object):Void {  
    switch (event.mouseIndex) {  
        ...  
    }  
}
```

```
btnInstance.setSize(100,22);  
btnInstance.width = 200;  
btnInstance.label = "text";  
btnInstance.toggle = true;  
btnInstance.selected = true;  
btnInstance.data = obj;  
invalidate() → draw();
```

63:

CLIK

UIComponent.invalidate()
UIComponent.draw()

(setter)
invalidate()

draw()

UIComponent.validateNow()

Scaleform CLIK

2

:

1.

2.

scale9grid

Flash

9

Flash

Flex

“

”

CLIK Scaleform 9
CLIK

Flash IDE
setSize()

width height

Flash IDE
LivePreview

MovieClip

CLIK Scaleform Player

Launcher

Scale9Grid
 MovieClip
 Player

2
 MovieClip

Scaleform

1
 MovieClip

scale9grid
 scale9grid
 Scale9grid

Scaleform Player
 Flash
 Flash

Constraints (gfx.utils.Constraints)

ScrollBar

Constraints

ScrollBar

configUI()

constraints

draw()

constraints
draw()

```

private function configUI():Void {
    ...
    constraints = new Constraints(this);
    // upArrow
    constraints.addElement(downArrow, Constraints.BOTTOM);
    constraints.addElenent(track, Constraints.TOP|Constraints.BOTTOM);
    ...
}

private function draw():Void {
    ...
    constraints.update(__width, __height);
    ...
}

```

dataProvider (gfx.interfaces.IDataProvider) dataProvider Scaleform CLIK IDataProvider API

dataProvider dataProvider

dataProvider CoreList (ScrollingList TileList)
OptionStepper DropdownMenu CLIK
IDataProvider IDataProvider
API

initialize() DataProvider (gfx.data.DataProvider)
dataProvider dataProvider ActionScript dataProvider

IDataProvider
dataProvider :

```
myComponent.dataProvider = [ "data1",
                              4.3,
                              {label:"anObjectElement", value:6} ];
```

dataProvider :

- *length* : getter
- *requestItemAt* : dataProvider OptionStepper
1
- *requestItemRange* : dataProvider
ScrollingList
- *indexOf* :
- *invalidate* : dataProvider
"dataChange" dataProvider

length

dataProvider

ActionScript 2

dataProvider

ExternalInterface.call

3

Scaleform CLIK
Scaleform
Tween

Flash 8 Tween
Tween
mx.transitions.easing.*

(gfx.motion.Tween)

CLIK Tween

Flash
Flash 8 MovieClip
CLIK

CLIK Tween onEnterFrame

MovieClip
MovieClip

1

MovieClip

MovieClip

Tween MovieClip 2 tweenTo() tweenFrom()
tweenTo() MovieClip
tweenFrom()

MovieClip
Tween Tween.init()

:

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init(); // MovieClip
// 1
myMovieClip.tweenTo(1, {_x: 200, _alpha: 0}, Strong.easeIn);
```


onTweenComplete()

```
import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();
mc.onTweenComplete = function() {
    trace("Tween has finished!");
}
mc.TweenTo(5, {_rotation: 20}, Bounce.easeOut);
```

Scaleform CLIK PopUpManager (gfx.managers.PopUpManager)
Dialog
PopUpManager

PopUpManager
createPopUp()
) ID MovieClip (CLIK context
relativeTo
MovieClip

```
import gfx.managers.PopUpManager;
PopUpManager.createPopUp(context, "MyToolTipLinkageID",
    {_x: 200, _y:200}, relativeTo);
```

Scaleform topmostLevel

SWF

topmostLevel

topmostLevel
Z

topmostLevel true

DragManager (gfx.managers.DragManager)

/

startDrag() MovieClip ID

stopDrag()
DragManager InputDelegate



64:

DragManager DragManager

Scaleform CLIK DragTarget (gfx.controls.DragTarget)
DragTarget UIComponent CLIK
DragManager

```

DragTarget
/
DragTarget
DragTarget
dragBegin dragEnd
DragTarget DragManager

```

```

CLIK
2
Inventory Demo
DragManager DragTarget
2 DragTarget
2 InventorySlot ItemSlot
CLIK

```

```

InventorySlot (gfx.controls.InventorySlot)
DragTarget DragManager

```

```

InventorySlot
ItemSlot (gfx.controls.ItemSlot) InventorySlot
ItemSlot DragTarget 2 CLIK Button
& ItemSlot

```

```

Delegate (gfx.utils.Delegate)
Scaleform CLIK
DragManager :

```

```

eventProvider.onMouseMove = Delegate.create(this, doDrag);

```

```

Locale (gfx.utils.Locale)
Scaleform
ActionScript

```

Button Label TextInput
Locale.getTranslatedString()

Locale

Scaleform CLIK

Scaleform CLIK

Michael	Jordan
Roger	Federer
Michael	Schumacher
Tiger	Woods
Babe	Ruth
Wayne	Gretzky

65: 2

ScrollingList

ScrollingList

ListItemRenderer

ListItemRenderer 1

textField

textField

textField

2 textField

2 textField

ListItemRenderer

ListItemRenderer

ScrollingList

2 textField

dataProvider

□

```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                    {fname: "Roger",   lname: "Federer"},
                    {fname: "Michael", lname: "Schumacher"},
                    {fname: "Tiger",   lname: "Woods"},
                    {fname: "Babe",    lname: "Ruth"},
                    {fname: "Wayne",   lname: "Gretzky"},
                    {fname: "Usain",   lname: "Bolt"}];
```

```

dataProvider      2      2      fname      lname
                2      2      textField
                ListItemRenderer  1      textField      2
textField
ListItemRenderer      MyItemRenderer
                ListItemRenderer      2
textField      (      FLA
                MyItemRenderer.as      :

```

```

import gfx.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

    public var textField1:TextField;
    public var textField2:TextField;

    public function MyItemRenderer() { super(); }

    public function setData(data:Object):Void {
        this.data = data;
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }

    private function updateAfterStateChange():Void {
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }
}

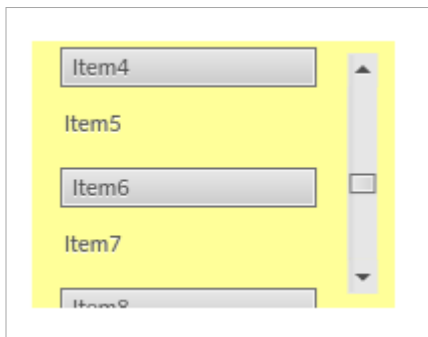
```

```

ListItemRenderer      setData      updateAfterStateChange
                setData
                (ScrollingList      TileList      )
                ListItemRenderer      1      textField
MyItemRenderer      textField      updateAfterStateChange
                ListItemRenderer
                textField
                dataProvider
                ListItemRenderer
                ListItemRenderer
                ListItemRenderer      'textField1'      'textField2'      2      textField

```

	MyItemRenderer	MyItemRenderer	
	itemRenderer		'ListItemRenderer'
'MyItemRenderer'			
FLA	dataProvider	fname	lname
2			



66: CLIK

CLIK ScrollBar

:

- *actions* : ActionScript
- *scrollbar* : CLIK ScrollBar
 'sb'
- *mask* : MovieClip
- *content* :
- *background* :

'content' y 0

code

ActionScript

:

```
// 133 ( )
sb.setScrollProperties(1, 0, content._height - 133);
sb.position = 0;

sb.addEventListener("scroll", this, "onScroll");
function onScroll() {
    content._y = -sb.position;
}
```

setScrollProperties

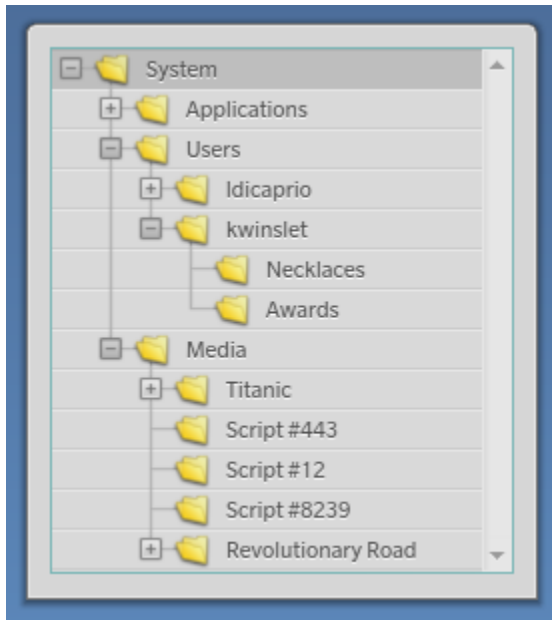
Scaleform

HUD
Scaleform

Scaleform CLIK

Flash ActionScript 2

ScrollingList TreeView



67: TreeView

TreeView Demo (*Resources/AS2/CLIK/demos/TreeViewDemo.fla*)
ListItemRenderer DataProvider ScrollingList

ScrollingList :

-
-

ListItemRenderer

-

ListItemRenderer

```
TreeViewItemRenderer
DataProvider
```

TreeViewDataProvider
TreeViewConstants

ActionScript 2

Object

•

•

...

2

Object

ScrollingList

TreeViewDataProvider DataProvider

ListItemRenderer

/

/

TreeViewDataProvider CLIK

TreeViewItemRenderer

ScrollingList

TreeViewDataProvider

ListItemRenderer

TreeViewItemRenderer

ListItemRenderer textField

TreeViewConstants

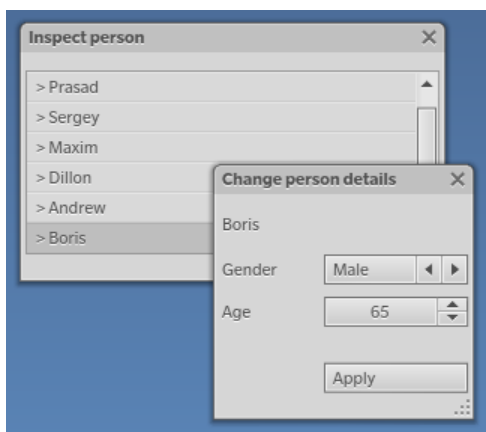
:

- :
 -
 -
- 2

ScrollingList

ScrollBar

/
ScrollingList



68: Window

Window Demo (*Resources/AS2/CLIK/demos/WindowDemo.fla*)

Window

Resources/AS2/CLIK/demos/com/scaleform

Window

UIComponent

CLIK

Window

Window

:

-
-
-
-
-

()

Window

:

Title	Window			
formPadding	Window			
formType	'symbol' 'swf'			
	'symbol'			
	'swf'	SWF		
formSource	)	SWF	(formType 'swf'	'symbol'
allowResize	/			
minWidth,	max			
maxWidth,	min			
minHeight,	max			
maxHeight	max 0			
	min			

:

```
[Inspectable(name="formType", enumeration="symbol,swf")
private var _formType:String = "symbol";
```

Inspectable
Inspectable
_formType

Flash
formType

getter setter

```
[Inspectable(defaultValue="Title")]  
public function get title():String { return _title; }  
public function set title(value:String):Void {  
    _title = value;  
    invalidate();  
}
```

Window

CLIK

Button

Window
2

UIComponent
constraints

configUI() configUI() draw()
draw()
constraints

configForm()

Flash 8 MovieClipLoader
Scaleform

MovieClipLoader.onLoadComplete()

configForm()

constraints

Window

draw()

constraints
constraints

validateNow()

UIComponent

draw()
draw()
UIComponent

validateNow()
Window Demo

draw()

validateNow()

SWF

Window

Window

2

```
private function onMouseDown() {  
    var targetObj:Object = Mouse.getTopMostEntity();  
    while (targetObj != null && targetObj != _root)  
    {  
        if (targetObj == this) {  
            swapDepths(_parent.getNextHighestDepth());  
            return;  
        }  
        targetObj = targetObj._parent;  
    }  
}
```

1

WindowManager

WindowManager

Window